Hooke’s law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect spring displacement with restoring force.

SUCCESS CRITERIA - I can...

1. Measure displacement from the natural length.
2. Use spring constant in N/m.
3. Point the spring force back toward equilibrium.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “spring constant” and “displacement” to explain your model or result. Sentence starters are optional.

spring constant: how much an ideal spring’s force changes per stretch

displacement: change from initial position to final position, with direction

equilibrium: zero net force and zero acceleration

01 Warm-up: bring back an earlier idea

A point-mass pendulum drops 0.6 m vertically from rest.
Use g=10 and neglect drag.
Find bottom speed.

Word help: pendulum = an object that swings about a fixed support

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $F_{spring} = -kx$; $|F_{spring}| = k|x|$.
x is displacement from natural length; k is spring constant. The force restores the spring, within its linear range.

Worked reasoning: A spring with k=100 N/m stretches 0.03 m, giving force magnitude kx=3 N. Its force on the attached object points toward equilibrium.

Your turn: Sketch equilibrium and stretched positions, then draw the restoring-force arrow.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Mark equilibrium and displacement x.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A spring is stretched right.
Draw the force it exerts on the attached block. What changes if compressed left?

Sentence starter: The … in my model represents … . Its label shows … .



Mark equilibrium and displacement x.

The zigzag is the spring; the rectangle is the attached block.

Marzano check - circle: ① ② ③

04 Apply the physics

A stationary 1 kg load stretches a vertical spring 0.050 m.
Use g = 9.8 N/kg.
Find the spring constant, then its force magnitude at a 0.12 m stretch.
Sketch and label the load’s forces for the stationary case.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.

Add and label force arrows.

Marzano check - circle: ① ② ③

05 Explain your reasoning

A student says k doubles when the stretch doubles.
Explain which quantity actually doubles for the same ideal spring.

Word help: ideal = using stated simplifying conditions

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A force parallel to motion does 24 J over 4 m.
Find its size. What work does the same force do with zero displacement?
Label force and displacement, including their relative direction.

Word help: parallel = along the same direction as a line or surface

Sentence starter: I found … [unit]. This value means … in this situation.

Add force and displacement arrows; label the angle.



Force: ____ N Displacement: ____ m

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

With right positive, k=150 N/m and x=-0.08 m, find signed spring force.
Mark the displacement from equilibrium and draw the restoring-force direction.

Sentence starter: I found … [unit]. This value means … in this situation.



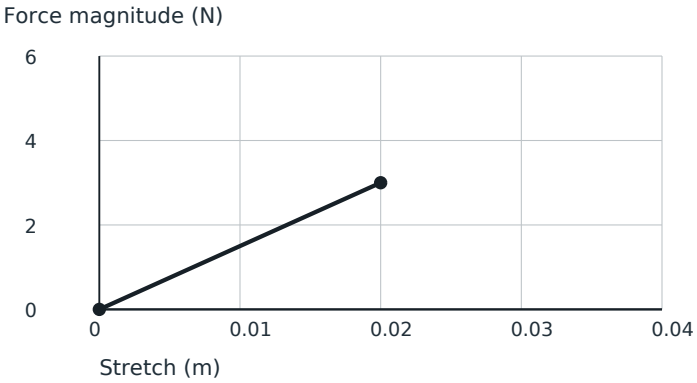
Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A spring graph passes through (x=0.02 m, F=3 N) and the origin.
Find k, then predict restoring force at displacement x=-0.04 m using F=-kx.
Mark the displacement from equilibrium and draw the restoring-force direction.

Sentence starter: I predict … . My model supports this because … .



Marzano check - circle: 1 2 3